

Guided Inquiry: Pressure Measurement and Manometry

Objective: Students will be able to apply the principles of fluid statics to determine the pressure of a gas using U-tube and inclined manometers. Students will also evaluate the sensitivity and precision advantages of different manometer configurations.

Prerequisites: Basic understanding of density, specific weight (γ), and trigonometry.

Part 1: The U-Tube Manometer

Open the Simulation: Locate the "U-Tube Manometer" section on the left side of the screen.

1. Initial State (Equilibrium)

- Set the **Tank Pressure** (P_{gas}) slider to **760 mmHg**.
- **Observe:** Look at the fluid levels in the left and right legs of the U-tube.
- **Q1:** What is the vertical height difference (Δh) between the two fluid columns? Explain why this occurs given that $P_{atm} = 760$ mmHg.

2. Pressure Application

- Increase the **Tank Pressure** to **780 mmHg**.
- **Observe:** Which direction does the fluid move?
- **Q2:** Using the concept of hydrostatic pressure ($P = P_{atm} + \gamma h$), explain why the fluid level drops on the tank side and rises on the open side.

3. Data Analysis

- Set the slider to three different values (e.g., 770, 785, 800 mmHg) and record the resulting Δh (Vertical Height Difference) displayed in the info box.

P_{gas} (mmHg)	P_{atm} (mmHg)	Measured Δh (mm)	Calculated difference ($P_{gas} - P_{atm}$)
770	760		
785	760		
800	760		

- **Q3:** Based on your data, write a mathematical equation relating P_{gas} , P_{atm} , and h .

Part 2: The Inclined Manometer

Open the Simulation: Locate the “Inclined Manometer” section on the right side of the screen.

Note: This simulation models a “Well-Type” manometer where the reservoir area is large enough that its level change is negligible.

4. Geometry and Reading

- Set the **Tank Pressure** to **780 mmHg**.
- **Observe:** Look at the value for **Vertical Δh** and the **Inclined Length (L)**.
- **Q4:** How does the Vertical Δh here compare to the Δh you measured in the U-Tube Part 1 for the same pressure?

5. Deriving the Inclination Factor

- The tube is inclined at an angle $\theta = 30^\circ$.
- **Q5:** Using trigonometry, derive the relationship between the inclined length L , the vertical height h , and the angle θ .
- **Q6:** Verify your formula using the data from the simulation (e.g., at 800 mmHg). Does $L = h / \sin(30^\circ)$?

Part 3: Critical Thinking & Sensitivity

6. Sensitivity Analysis

- **Definition:** *Sensitivity* in this context is the change in scale reading per unit change in pressure ($\Delta L / \Delta P$).
- **Experiment:** Move the slider from 760 to 761 mmHg (a 1 mmHg change).
 - How much did the reading change on the U-Tube? (~ 1 mm)
 - How much did the reading change on the Inclined Manometer? (~ 2 mm)
- **Q7:** Which manometer provides a more precise reading for very small pressure changes? Why?

- **Q8 (Engineering Application):** If you needed to measure low-speed air currents (generating very small pressure differences) in an HVAC duct, would you choose a vertical U-tube or an inclined manometer? Explain your choice.